Macroeconomics #4

Overlapping Generations Model

Quang-Thanh Tran

RCPD, Tohoku University

CASED Pre-Master Macroeconomics
October 30, 2026

Today's lecture

- OLG Structure
- Competitive Equilibrium
- Dynamic Efficiency
- Multiple Equilibria
- Further Issues

Introduction and Motivation

Why Overlapping Generations?

In the Ramsey model:

- A single infinitely lived representative agent.
- Competitive equilibrium is always Pareto optimal.
- Dynamic efficiency is guaranteed.

But in reality:

- People are born, grow old, and die — finite lifetimes.
- Different generations coexist at each point in time.
- Young and old have different economic roles (working vs. retired).
- This heterogeneity has profound implications for efficiency and policy.

The OLG framework

The Overlapping Generations (OLG) model, due to Samuelson (1958) and Diamond (1965):

- Agents live for a finite number of periods (typically 2).
- At each date, a young generation and an old generation coexist.
- Young agents work, consume, and save.
- Old agents consume their savings and returns, then exit.

Key results that differ from Ramsey:

- Competitive equilibria can be dynamically inefficient (over-accumulation).
- The First Welfare Theorem may fail.
- Government policies (debt, social security) can improve welfare.

References

- Samuelson, P. A. (1958). "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money." *JPE*, 66(6), 467–482.
- Diamond, P. A. (1965). "National Debt in a Neoclassical Growth Model." *AER*, 55(5), 1126–1150.
- Barro, R. J. & Sala-i-Martin, X. (2004). *Economic Growth*, 2nd ed. MIT Press. Chapter 9.
- De la Croix, D. & Michel, P. (2002). *A Theory of Economic Growth: Dynamics and Policy in Overlapping Generations*. Cambridge University Press.

Environment

Generational structure

	Date 1	Date 2	Date 3	$\cdots$	Date t	Date $t+1$
Gen. 0	old					
Gen. 1	young	old				
Gen. 2		young	old			
$\vdots$				$\ddots$		
Gen. t					young	old

- At each date t, two generations coexist: generation t (young) and generation $t-1$ (old).
- Each generation lives for exactly two periods.
- The initial old (generation 0) live only in period 1.

Endowments and consumption

Endowments:

- Labor endowment: $(l_y, l_o) = (1, 0)$ for all generations.
  - Young supply 1 unit of labor inelastically.
  - Old do not work.
- Initial old are endowed with K_1 units of capital in the aggregate.
- Wage rate w_t and rental price r_t.

Consumption:

- Initial old: d_1.
- Generation t: (c_t, d_{t+1}) — consumption when young and when old.

Preferences

Initial old (generation 0): $u(d_1)$.

Generation t **(**$t \geq 1$**):**

$$U^t(c^t) = u(c_t) + \beta\, u(d_{t+1})$$

where:

- $u : \mathbb{R}_+ \to \mathbb{R}$ satisfies standard assumptions: increasing, concave, Inada conditions.
- $\beta \in (0, 1)$ is the discount factor.
- No utility from leisure: all labor endowment is supplied.
- No altruism or bequest motive.

Technology and goods

Production:

$$Y_t = F(K_t, N_t)$$

- F exhibits constant returns to scale (neoclassical).
- In per-capita terms: $y_t = f(k_t)$ where $k_t = K_t/N_t$.

Goods:

- The single good can be consumed or saved (invested as productive capital).
- Consumers own the capital.
- Capital depreciates at rate δ.
- Firms operate in perfect competition.

Demography

Population of generation t (young at date t):

$$N_t = N_0(1+n)^t, \qquad n > -1$$

- n is the population growth rate.
- At each date t: N_t young + N_{t-1} old.
- Total population at t: $N_t + N_{t-1} = N_t\left(1 + \frac{1}{1+n}\right)$.

Feasible allocations and the resource constraint

A feasible allocation is a non-negative sequence of consumption $(d_1, \{c^t\}_{t=1}^{\infty})$ and capital $\{K_{t+1}\}_{t=1}^{\infty}$ satisfying the resource constraint for each period t:

$$N_t c_t + N_{t-1} d_t + K_{t+1} - (1-\delta)K_t \leq F(K_t, N_t)$$

In per-capita (of young) terms:

$$c_t + \frac{d_t}{1+n} + (1+n)k_{t+1} - (1-\delta)k_t \leq f(k_t)$$

Market structure

Households:

- Buy goods, which they either consume or save.
- Supply labor to firms when young.
- Own the capital, which they rent to firms.

Firms:

- Operate in perfect competition.
- Rental rate is r_t and price of goods is 1.
- Wage rate is w_t.

Timing of events

In period t:

1. Production takes place using labor of generation t and capital saved by generation $t-1$.
2. Generation t receives wage w_t.
3. Generation t decides consumption c_t and saving s_t^t.
4. Savings become physical capital.

In period $t+1$:

1. Production uses labor of generation $t+1$ and capital saved by generation t.
2. Generation t (now old) receives return $(1 + r_{t+1} - \delta)\, s_t^t$.
3. Old consume and exit the economy.

Decision Problems

Initial old (generation 0)

The initial old simply maximize:

$$\max_{d_1} \; u(d_1) \qquad \text{s.t.} \quad d_1 \leq (1 + r_1 - \delta) \frac{K_1}{N_0}$$

Since u is increasing, the constraint binds:

$$d_1 = (1 + r_1 - \delta) \frac{K_1}{N_0}$$

Since the initial old live for only one period, they consume all of their capital income: technically there is no choice to make.

Generation t: the two-period problem

Generation t solves:

$$\max_{\{c_t,\, s_t^t,\, d_{t+1}\}} \quad u(c_t) + \beta\, u(d_{t+1})$$

subject to:

$$\begin{aligned} c_t + s_t^t &\leq w_t && \text{(young budget)} \\ d_{t+1} &\leq (1 + r_{t+1} - \delta)\, s_t^t && \text{(old budget)} \end{aligned}$$

Combining the two budget constraints yields the lifetime budget constraint:

$$c_t + \frac{d_{t+1}}{1 + r_{t+1} - \delta} = w_t$$

Euler equation in the OLG

Forming the Lagrangian and taking FOCs:

$$\frac{u'(c_t)}{\beta\, u'(d_{t+1})} = 1 + r_{t+1} - \delta \equiv R_{t+1}$$

Interpretation:

- The MRS between young and old consumption equals the gross return.
- Same structure as the Ramsey Euler equation, but agents live only two periods.

From the Euler equation, we can express saving as a function of wages and returns:

$$s_t = s(w_t, R_{t+1})$$

via the implicit function theorem.

Using the implicit function theorem:

$$\frac{\partial s}{\partial w_t} = \frac{u''(c_t)}{u''(c_t) + \beta R_{t+1}^2 u''(d_{t+1})} \in (0,1)$$

Higher wages $\Rightarrow$ more saving (unambiguous).
Higher returns on savings $R_{t+1} \Rightarrow$ more savings?

$$\frac{\partial s}{\partial R_{t+1}} \begin{cases} > 0 & \text{if } \theta < 1 \\ = 0 & \text{if } \theta = 1 \\ < 0 & \text{if } \theta > 1 \end{cases}$$

$\theta = -\frac{c\,u''(c)}{u'(c)}$: inverse of the intertemporal elasticity of substitution (IES $= 1/\theta$).

- When $\theta < 1$, high IES: agent willing to shift consumption over time, substitute dominates. Higher returns make saving attractive.
- When $\theta > 1$, low IES: agent dislike jumps, wants consumption smoothing, income dominates. Higher returns mean agents can save less today and still enjoy enough consumption when old.
- When $\theta = 1$, substitution and income effects offset each other.

- Substitution effect: $c \downarrow, s \uparrow$
  A higher interest rate raises the reward for postponing consumption. The relative price of future consumption falls, inducing the agent to save more at the expense of current consumption.
- Income effect: $c \uparrow, s \downarrow$
  A higher interest rates increases the return on saving, so the same amount of saving yields higher future consumption, making lifetime income increase. This positive wealth effect incentivizes agents to consume more in both periods. The increase in present consumption reduces saving.

Firms' problem

Each firm solves:

$$\max_{K_t^d, N_t^d} \; F(K_t^d, N_t^d) - r_t K_t^d - w_t N_t^d$$

First-order conditions:

$$r_t = F_K(K_t^d, N_t^d) = f'(k_t)$$

$$w_t = F_N(K_t^d, N_t^d) = f(k_t) - k_t f'(k_t)$$

- Factors are paid their marginal products.
- Equilibrium profits are zero (CRS).

Definition of competitive equilibrium

Definition. A competitive equilibrium with production given K_1 is a sequence of factor prices $\{r_t, w_t\}_{t=1}^{\infty}$ and allocations for households $(d_1, \{c^t, s_t^t\}_{t=1}^{\infty})$ and firms $\{K_t^d, N_t^d\}_{t=1}^{\infty}$ such that:

1. Given prices, d_1 solves the initial old's problem and (c^t, s_t^t) solves generation t's problem.
2. Given prices, (K_t^d, N_t^d) solves the firm's problem.
3. Markets clear:

$$\begin{aligned} \text{Goods: } & N_t c_t + N_{t-1} d_t + K_{t+1} - (1-\delta) K_t = F(K_t^d, N_t^d) \\ \text{Labor: } & N_t = N_t^d \\ \text{Capital: } & N_{t-1} s_{t-1}^{t-1} = K_t^d \end{aligned}$$

Walras's Law

Aggregating the two budget constraints for generations $t-1$ and t at period t:

$$N_t c_t + N_t s_t^t + N_{t-1} d_t \leq N_t w_t + (1 + r_t - \delta) N_{t-1} s_{t-1}^{t-1}$$

Using capital market clearing ($N_{t-1} s_{t-1}^{t-1} = K_t$) and $K_{t+1} = N_t s_t^t$:

$$N_t c_t + N_{t-1} d_t + K_{t+1} - (1 - \delta) K_t \leq w_t N_t + r_t K_t = F(K_t^d, N_t^d)$$

The goods market clears automatically if factor markets and budget constraints are satisfied (Walras's Law).

Equilibrium law of motion

From optimization, saving is $s_t^t = s(w_t, R_{t+1})$, with $R_{t+1} = 1 + r_{t+1} - \delta$.
Capital market clearing: $N_t s_t^t = K_{t+1}$.
In per-capita terms:

$$(1+n)\, k_{t+1} = s(w_t, R_{t+1})$$

Using $w_t = f(k_t) - k_t f'(k_t)$ and $R_{t+1} = 1 + f'(k_{t+1}) - \delta$:

$$k_{t+1} = \frac{s\big(f(k_t) - k_t f'(k_t),\ 1 + f'(k_{t+1}) - \delta\big)}{1+n}$$

This is an (implicit) first-order difference equation in k_t.

Given $k_1 = K_1/N_1$, the entire equilibrium path is determined.

Steady state

Definition. A steady state is a value k^{ss} such that $k_{t+1} = k_t = k^{ss}$:

$$(1+n)\,k^{ss} = s\big(f(k^{ss}) - k^{ss} f'(k^{ss}),\ 1 + f'(k^{ss}) - \delta\big)$$

At the steady state, all variables are constant:

$$r^{ss} = f'(k^{ss}), \qquad R^{ss} = 1 + r^{ss} - \delta, \qquad w^{ss} = f(k^{ss}) - k^{ss} f'(k^{ss})$$
$$s^{ss} = (1+n)\,k^{ss}, \qquad c^{ss} = w^{ss} - s^{ss}, \qquad d^{ss} = R^{ss}\,s^{ss}$$

Unlike Ramsey, the steady state is not necessarily unique — multiple steady states are possible depending on $s(\cdot)$.

Example: Log Utility and Cobb–Douglas

Setup

Assume:

$$U(c_t, d_{t+1}) = \log(c_t) + \beta \log(d_{t+1}), \qquad F(K, N) = AK^{\alpha} N^{1-\alpha}$$

Lagrangian:

$$\mathcal{L} = \log(c_t) + \beta \log(d_{t+1}) + \lambda_t \left(w_t - c_t - \frac{d_{t+1}}{1 + r_{t+1} - \delta} \right)$$

FOCs:

$$c_t : \quad \frac{1}{c_t} = \lambda_t$$

$$d_{t+1} : \quad \frac{\beta}{d_{t+1}} = \frac{\lambda_t}{1 + r_{t+1} - \delta}$$

Combining: $d_{t+1} = \beta(1 + r_{t+1} - \delta)\, c_t$.

Solving for consumption and saving

Using the budget constraint:

$$c_t = \frac{w_t}{1+\beta}, \qquad d_{t+1} = \frac{\beta(1+r_{t+1}-\delta)\, w_t}{1+\beta}$$

$$s_t^t = \frac{\beta}{1+\beta}\, w_t$$

- With log utility, saving is a constant fraction of wage income.
- Saving does not depend on the interest rate (income and substitution effects exactly cancel).
- Higher β (more patience) $\Rightarrow$ higher saving rate.

Equilibrium law of motion

From the firm's problem: $r_t = A\alpha k_t^{\alpha-1}$, $w_t = A(1-\alpha)k_t^{\alpha}$.
Capital market clearing: $(1+n)k_{t+1} = s_t = \frac{\beta}{1+\beta} w_t$:

$$k_{t+1} = \frac{1}{1+n} \cdot \frac{\beta}{1+\beta} A(1-\alpha)\, k_t^{\alpha}$$

- This is a concave function of k_t (since $\alpha < 1$).
- Unique positive steady state, globally stable.

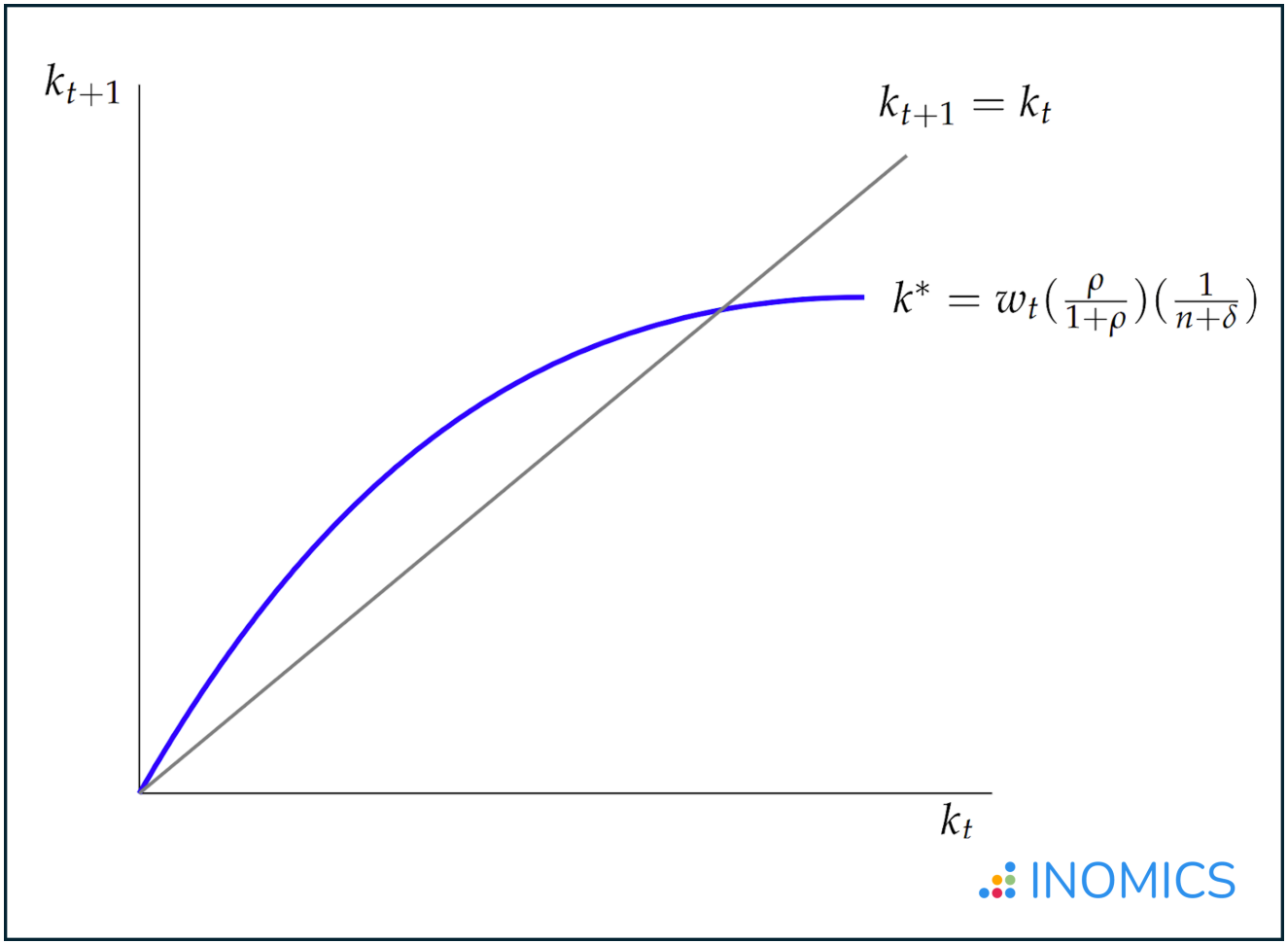


Figure: OLG dynamics with Cobb-Douglas production function and log utility

Steady-state values

Setting $k_{t+1} = k_t = k^{ss}$:

$$k^{ss} = \left(\frac{\beta A(1-\alpha)}{(1+\beta)(1+n)} \right)^{\frac{1}{1-\alpha}}$$

Other steady-state values:

$$r^{ss} = \frac{\alpha\,(1+n)\,(1+\beta)}{\beta\,(1-\alpha)}$$

$$w^{ss} = A(1-\alpha) \left(\frac{\beta A(1-\alpha)}{(1+\beta)(1+n)} \right)^{\frac{\alpha}{1-\alpha}}$$

$$c^{ss} = \frac{w^{ss}}{1+\beta}, \qquad d^{ss} = (1 + r^{ss} - \delta) \frac{\beta\, w^{ss}}{1+\beta}$$

Comparative statics

From $k^{ss} = \left(\frac{\beta A(1-\alpha)}{(1+\beta)(1+n)}\right)^{1/(1-\alpha)}$:

- Higher TFP ($A \uparrow$): $k^{ss} \uparrow$, $w^{ss} \uparrow$, $c^{ss} \uparrow$, $d^{ss} \uparrow$.
- Higher patience ($\beta \uparrow$): $k^{ss} \uparrow$ (more saving).
- Higher population growth ($n \uparrow$): $k^{ss} \downarrow$ (capital dilution).

Multiple Equilibria

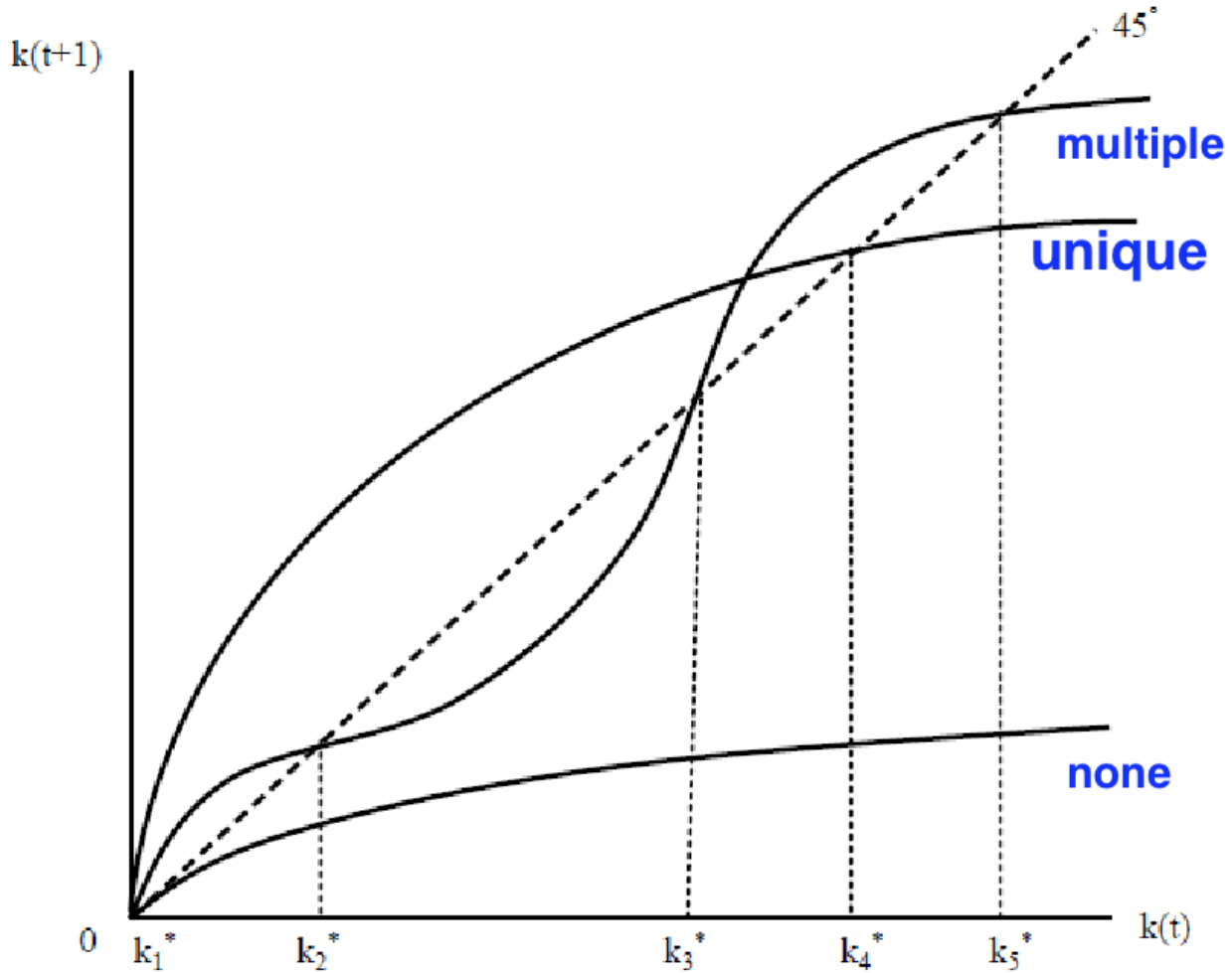


Figure: Steady state equilibrium: unique, two stable, no steady state

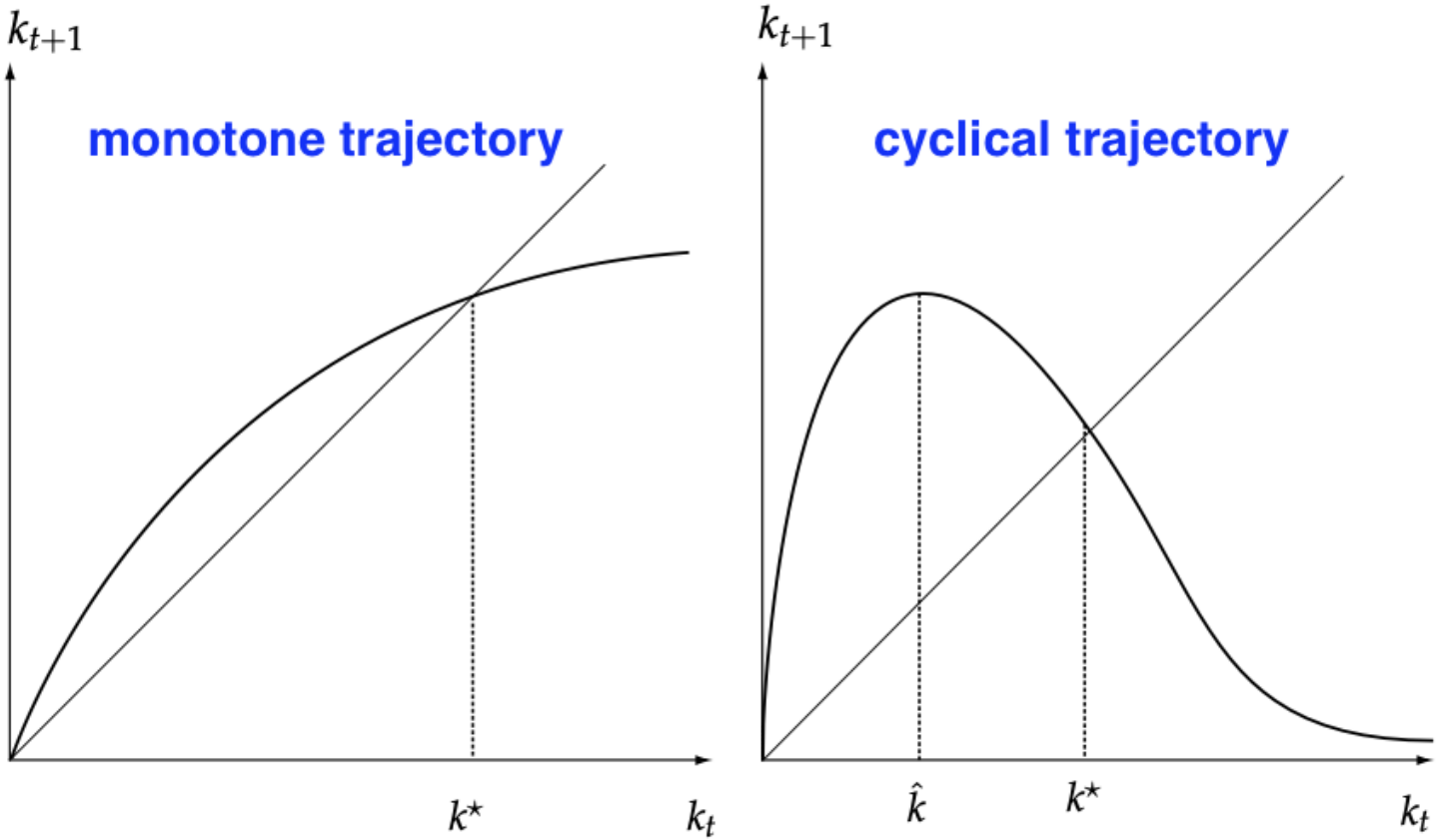


Figure 1.16. Rational and myopic dynamics for $\gamma > \alpha$. The two dynamics with perfect and myopic foresight have the same positive steady state $k^\star$ (which is unique with a Cobb–Douglas production function and a CIES utility function). $k^\star$ is stable for the rational dynamics, and the trajectory is monotonic. $k^\star > \hat{k}$ can be stable or unstable for the myopic dynamics, and the trajectory is oscillating near $k^\star$.

Figure: Different patterns of convergence to the steady state

Dynamic Efficiency and Over-Accumulation

Is the competitive equilibrium Pareto optimal?

Question: Is the steady state of the competitive equilibrium efficient?

To answer this, solve the Social Planner's problem:

$$\max_{\{d_1, \{c^t, k_{t+1}\}_{t=1}^{\infty}\}} \beta \log(d_1) + \sum_{t=1}^{\infty} [\log(c_t) + \beta \log(d_{t+1})]$$

subject to the resource constraint:

$$c_t + \frac{d_t}{1+n} + (1+n)k_{t+1} - (1-\delta)k_t = Ak_t^{\alpha} \qquad \forall\, t$$

Lagrangian

$$\mathcal{L} = \beta \log d_1 + \sum_{t=1}^{\infty} \Bigg[\log c_t + \beta \log d_{t+1} - \lambda_t \Big(c_t + \frac{d_t}{1+n} + (1+n)k_{t+1} - (1-\delta)k_t - Ak_t^{\alpha} \Big) \Bigg]$$

Planner's FOCs

Taking FOCs of the Lagrangian:

$$\begin{aligned} c_t: &\quad \frac{1}{c_t} = \lambda_t \\ d_{t+1}: &\quad \frac{\beta}{d_{t+1}} = \frac{\lambda_{t+1}}{1+n} \\ k_{t+1}: &\quad (1+n)\lambda_t = \lambda_{t+1}\left(A\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right) \end{aligned}$$

Combining the three conditions:

$$\begin{aligned} d_{t+1} &= \beta(1+n)c_{t+1}, \\ \frac{1}{c_t} &= \frac{\beta}{d_{t+1}}\left(A\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right) \end{aligned}$$

implying

$$(1+n)c_{t+1} = c_t\left(A\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right)$$

Planner's steady state

At the planner's steady state, $c_t = c_{t+1}$ holds. This yields:

$$k^* = \left(\frac{A\alpha}{n+\delta}\right)^{\frac{1}{1-\alpha}}$$

This is the Golden Rule capital stock: $f'(k^*) - \delta = n$ maximizes steady-state $c + \frac{d}{1+n} = f(k) - (n+\delta)k$.

Compare with the competitive equilibrium:

$$k^{ss} = \left(\frac{\beta A(1-\alpha)}{(1+\beta)(1+n)}\right)^{\frac{1}{1-\alpha}}$$

Is k^{ss} above or below k^*? It depends on parameters!

Over-accumulation: a numerical example

Set $\delta = n = A = 1$, $\alpha = 0.3$, $\beta = 1$:

- Planner: $k^* = \left(\frac{0.3}{2}\right)^{1/0.7} = (0.15)^{10/7} \approx 0.067$.
- Market: $k^{ss} = \left(\frac{1 \cdot 0.7}{2 \cdot 2}\right)^{1/0.7} = (0.175)^{10/7} \approx 0.083$.

Since $k^{ss} > k^*$:

- The competitive equilibrium has too much capital.
- The net return $r^{ss} - \delta = \frac{0.3 \cdot 2 \cdot 2}{0.7} - 1 \approx 0.71 < n = 1$: the economy is dynamically inefficient.
- The steady state is not Pareto optimal.

This cannot happen in the Ramsey model! There the modified Golden Rule $f'(k) - \delta = \rho + \sigma g$, together with the TVC condition $\rho + \sigma g > n + g$ (σ = CRRA coefficient, g = technology growth), rules out over-accumulation.

Intuition for over-accumulation

There are two ways to provide consumption for the old:

1. **Capital saving**: give up 1 unit when young, invest it, receive $1 + r - \delta$ when old.
2. **Direct transfer**: each young gives 1 unit to the old. Since population grows at rate n, there are $1 + n$ young for every old, thus, each old receives $1 + n$ units.

▷ Say $n = 0.5$ (population grows by 50% every period), $r - \delta = 0.2$

1. If the young agent saves 1 unit, he gets $(1 + r - \delta) = 1.2$ units when old.
2. If the old agent receives 1 unit from current young, he gets $1 + n = 1.5$ units.

The transfer yields 25% more ($1.5/1.2 = 1.25$) and requires no capital at all.

The market cannot coordinate these transfers (strangers cannot commit future generations to a gift scheme), so it over-relies on capital. Too much saving drives r below $n \Rightarrow$ dynamic inefficiency.

Government Policy

Why government intervention?

Since the market equilibrium may exhibit over-accumulation, there is room for government intervention to improve welfare.

We consider two policies:

1. Government debt
2. Pay-as-you-go (PAYG) social security

Both work by reducing private saving and hence the capital stock, moving the economy closer to the Golden Rule.

Government debt: setup

- The government issues debt B_t held by domestic citizens.
- Debt-labor ratio: $b_t \equiv B_t/N_t = b$ (constant).
- Bonds have 1-period maturity and pay the same return as capital: $1 + r_t - \delta$ (no arbitrage; otherwise only one asset would be held)
- The government taxes the young with lump-sum tax τ to finance interest payments.

Government budget constraint:

$$B_t(1 + r_t - \delta) = B_{t+1} + N_t\tau_t$$

Dividing by N_t and using $B_{t+1}/N_t = (1 + n)b$, with constant b:

$$\tau_t = (r_t - \delta - n)\, b$$

If $r_t - \delta < n$ (dynamic inefficiency), $\tau_t < 0$: the debt is rolled over and the young are subsidised.

Equilibrium with government debt

The household's budget changes only for the young:

$$c_t + s_t^t = w_t - \tau_t$$

The saving function becomes: $s_t^t = s(w_t - \tau_t, R_{t+1})$.

Asset market clearing: savings must equal capital *plus* bonds:

$$N_t s_t^t = K_{t+1} + B_{t+1} \quad \Rightarrow \quad k_{t+1} = \frac{s(w_t - \tau_t, R_{t+1})}{1+n} - b$$

Government debt: log-utility example

With $U = \log c + \beta \log d$ and $F = AK^{\alpha}N^{1-\alpha}$:

$$c_t = \frac{w_t - \tau}{1+\beta}, \qquad s_t^t = \frac{\beta(w_t - \tau)}{1+\beta}$$

Law of motion:

$$k_{t+1} = \frac{1}{1+n} \cdot \frac{\beta}{1+\beta}\Big[A(1-\alpha)k_t^{\alpha} - (A\alpha k_t^{\alpha-1} - \delta - n)b\Big] - b$$

In steady state:

$$\hat{k}^{ss} = \frac{\beta}{(1+\beta)(1+n)}\Big[A(1-\alpha)(\hat{k}^{ss})^{\alpha} - (A\alpha(\hat{k}^{ss})^{\alpha-1} - \delta - n)b\Big] - b$$

Effect of increasing government debt

- Higher $b \Rightarrow$ lower $\hat{k}^{ss}$.
- If the economy was dynamically inefficient ($k^{ss} > k^{gold}$), moderate debt can improve welfare by reducing over-accumulation.
- If debt is too large, it can push $\hat{k}^{ss}$ below the Golden Rule — under-accumulation.

Numerical example

- Recall $\delta = n = A = 1$, $\alpha = 0.3$, $\beta = 1$:
  - Planner: $k^{GR} = \left(\frac{0.3}{2}\right)^{1/0.7} = (0.15)^{10/7} \approx 0.067$.
  - Market: $k^{ss} = \left(\frac{1 \cdot 0.7}{2 \cdot 2}\right)^{1/0.7} = (0.175)^{10/7} \approx 0.083$.
- By setting $b \approx 0.0111$, we have $k^{ss} \to k^{GR}$.
- At the Golden Rule $r - \delta = n$, so the tax is $\tau = (r - \delta - n)b = 0$: the debt is simply rolled over.

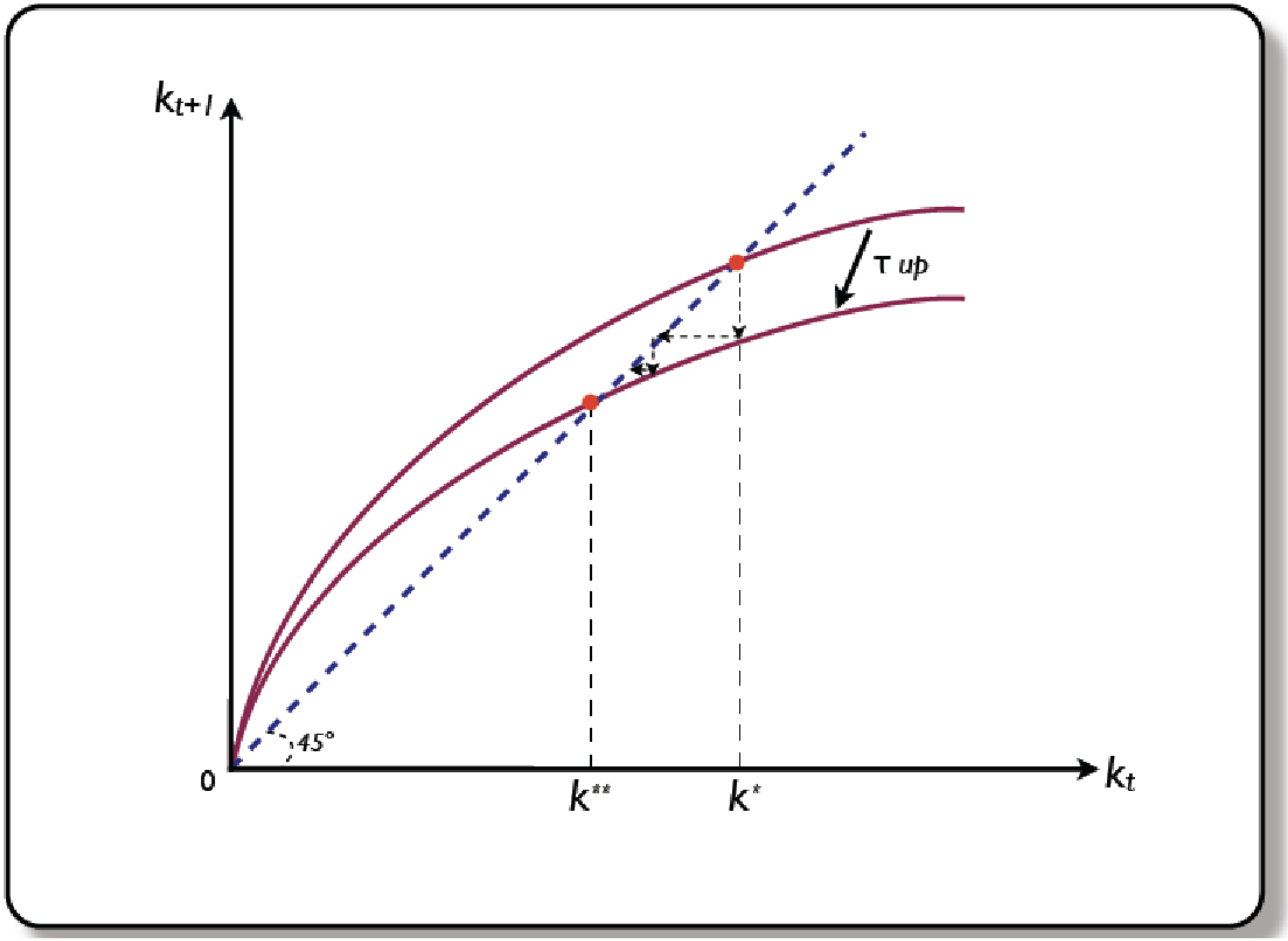


Figure: Effect of debt

PAYG social security: setup

In each period, the government:

- Taxes each young person τ_y.
- Transfers to each old person τ_o.

$$N_t \tau_y = N_{t-1} \tau_o$$

or

$$\tau_o = (1+n)\,\tau_y$$

Budget constraints:

$$\begin{aligned} c_t + s_t^t &\leq w_t - \tau_y \\ d_{t+1} &\leq (1 + r_{t+1} - \delta)\, s_t^t + \tau_o \end{aligned}$$

The old receive a transfer funded by the young a.k.a public pension system.

PAYG Example

With $U(c,d) = \log(c) + \beta \log(d)$, $F = AK^{\alpha}N^{1-\alpha}$:
Household problem:

$$\max_{s_t} \ \log(w_t - \tau_y - s_t) + \beta \log((1 + r_{t+1} - \delta)s_t + \tau_o)$$

FOC:

$$\frac{1}{w_t - \tau_y - s_t} = \beta \frac{1 + r_{t+1} - \delta}{(1 + r_{t+1} - \delta)s_t + \tau_o}$$

Savings:

$$s_t^t = \frac{\beta}{1+\beta}(w_t - \tau_y) - \frac{1}{1+\beta}\frac{\tau_o}{1 + r_{t+1} - \delta}$$

Equilibrium:

$$(1+n)k_{t+1} = \frac{\beta}{1+\beta}(A(1-\alpha)k_t^{\alpha} - \tau_y) - \frac{(1+n)\tau_y}{(1+\beta)(1 + A\alpha k_{t+1}^{\alpha-1} - \delta)}$$

In the steady state

$$(1+n)k = \frac{\beta}{1+\beta}\left(A(1-\alpha)k^{\alpha} - \tau_y\right) - \frac{(1+n)\tau_y}{(1+\beta)\left(1+A\alpha k^{\alpha-1} - \delta\right)}$$

We want to solve for τ_y so that $k = k^{GR}$.

Numerical Example

- Recall $\delta = n = A = 1$, $\alpha = 0.3$, $\beta = 1$:
  - Planner: $k^{GR} = \left(\frac{0.3}{2}\right)^{1/0.7} = (0.15)^{10/7} \approx 0.067$.
  - Market: $k^{ss} = \left(\frac{1 \cdot 0.7}{2 \cdot 2}\right)^{1/0.7} = (0.175)^{10/7} \approx 0.083$.
- By setting $\tau_y \approx 0.0222$, we have $k^{ss} \to k^{GR}$ (note $\tau_y = (1+n)b$: PAYG and debt are equivalent here).

Quantitative OLG
Beyond 2 periods

6-period OLG: Demographics and preferences

- Agents live for 6 periods:
  - Work: $s = 1, \ldots, 4$ (earn wage, pay tax, save).
  - Retire: $s = 5, 6$ (receive benefit b_t, consume savings).
- Lifetime utility evaluated at age 1:

$$\sum_{s=1}^{6} \beta^{s-1} u(c^s_{t+s-1}, l^s_{t+s-1})$$

- Period utility (CRRA with additive consumption term ψ and leisure):

$$u(c, l) = \frac{(c + \psi)^{1-\eta} l^{\gamma(1-\eta)} - 1}{1 - \eta}$$

- Endogenous labor supply: time endowment $l + n = 1$.
- Retirement: $l^5 = l^6 = 1$ (full leisure, $n^5 = n^6 = 0$).

6-period OLG: Budget constraints

Working periods ($s = 1, \ldots, 4$)**:**

$$k_{t+1}^{s+1} = (1 + r_t)\, k_t^s + (1 - \tau_t)\, w_t\, n_t^s - c_t^s$$

- Income: return on assets + after-tax wage.
- Outflow: consumption + next-period asset holdings.

Retirement periods ($s = 5, 6$)**:**

$$k_{t+1}^{s+1} = (1 + r_t)\, k_t^s + b_t - c_t^s$$

- No labor income; receive flat benefit b_t from government.

Boundary conditions:

$$k_t^1 = 0 \ \text{(born with no assets)}, \qquad k_t^7 = 0 \ \text{(no bequests)}$$

6-period OLG: Production, government, aggregation

Production (Cobb–Douglas):

$$Y_t = K_t^{\alpha} N_t^{1-\alpha}$$

$$w_t = (1-\alpha)\, K_t^{\alpha} N_t^{-\alpha}$$

$$r_t = \alpha\, K_t^{\alpha-1} N_t^{1-\alpha} - \delta$$

Government (balanced budget):

$$\tau_t\, w_t\, N_t = \frac{2}{6}\, b_t$$

Tax revenue from 4 working cohorts funds benefits for 2 retired cohorts. **Aggregation across cohorts ($\frac{1}{6}$ weight each):**

$$N_t = \frac{1}{6}\sum_{s=1}^{4} n_t^s, \qquad K_t = \frac{1}{6}\sum_{s=1}^{6} k_t^s$$

6-period OLG: Market clearing

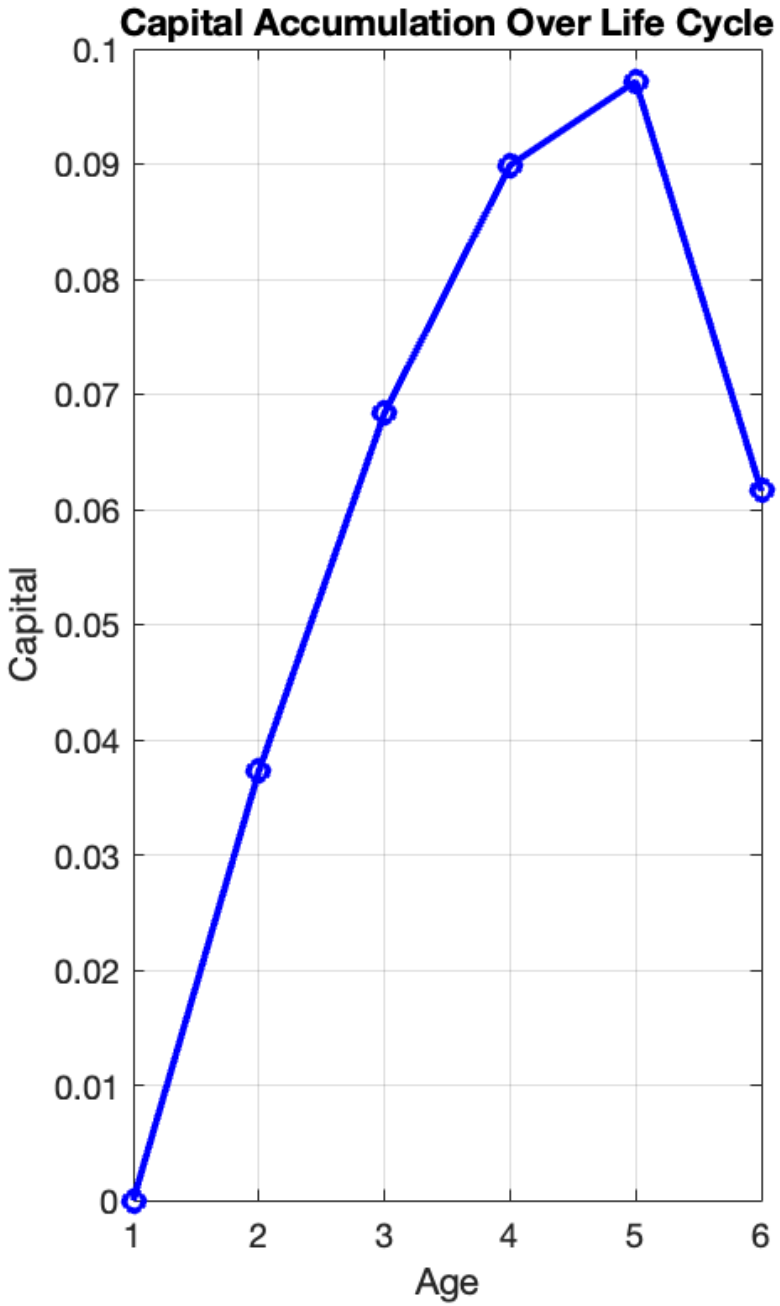
Goods market clearing:

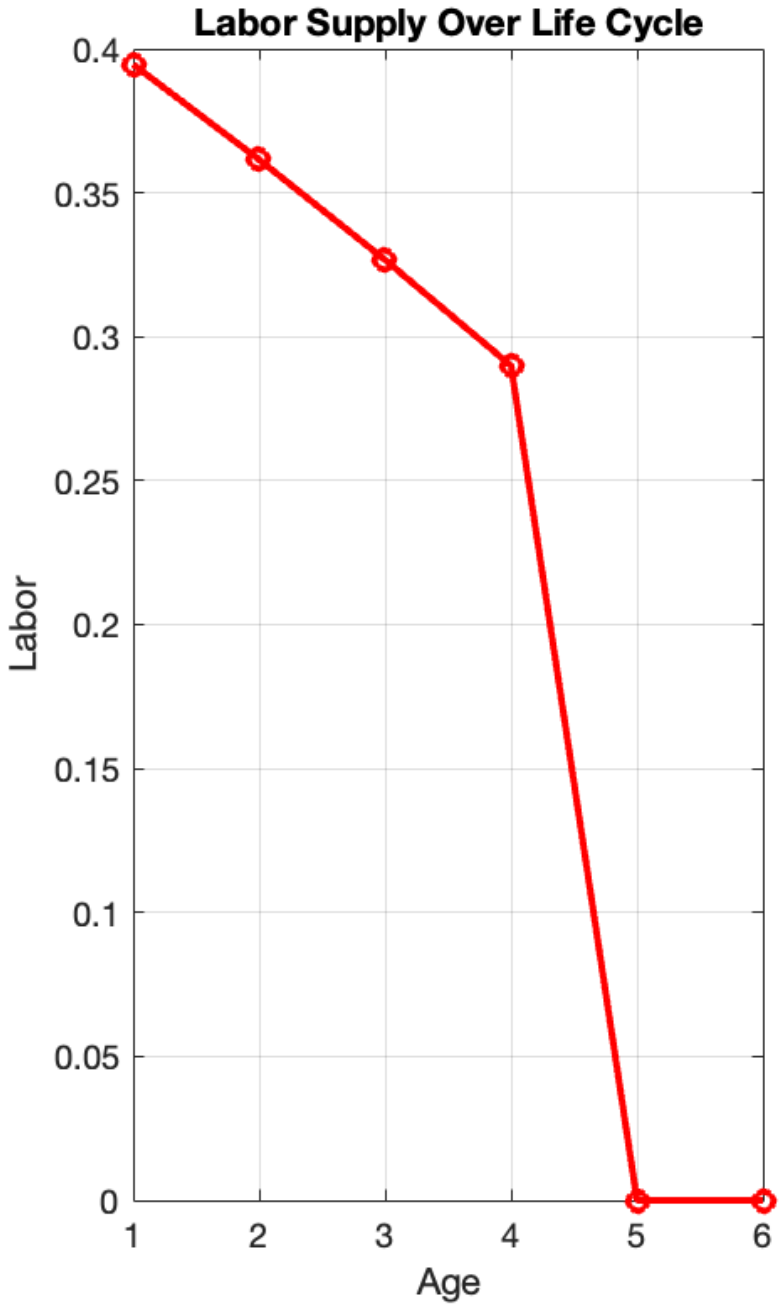
$$Y_t = \frac{1}{6}\sum_{s=1}^{6} c_t^s \; + \; K_{t+1} \; - \; (1-\delta)K_t$$

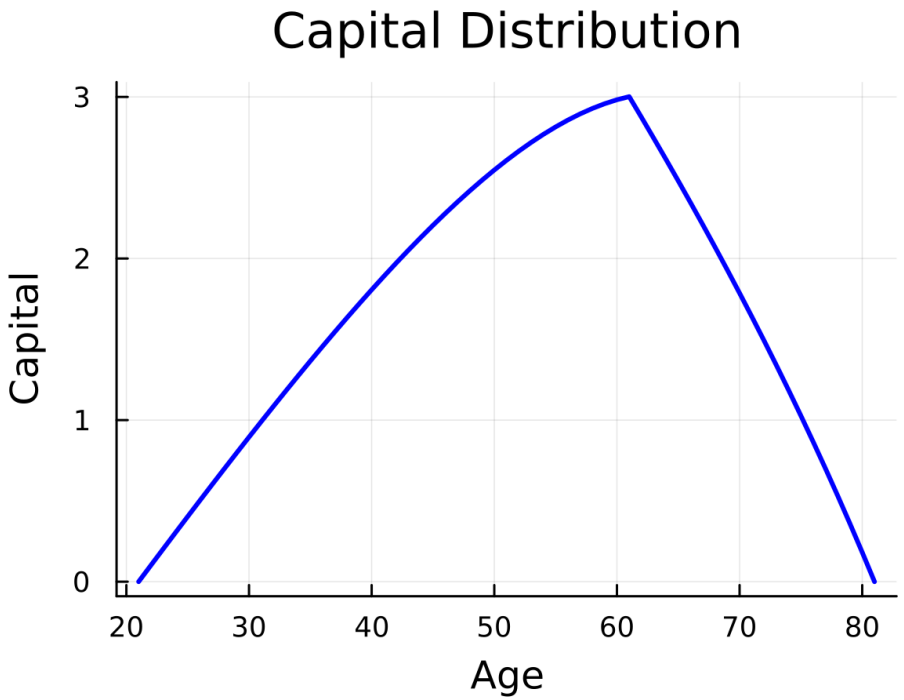
Total output = aggregate consumption (across all 6 cohorts) + net investment. **Factor markets:**

- Labor: supplied by cohorts $s = 1, \ldots, 4$, demanded by firms at rate w_t.
- Capital: savings of all cohorts rented to firms at rate r_t.

Walras's Law ensures consistency of the three market-clearing conditions.

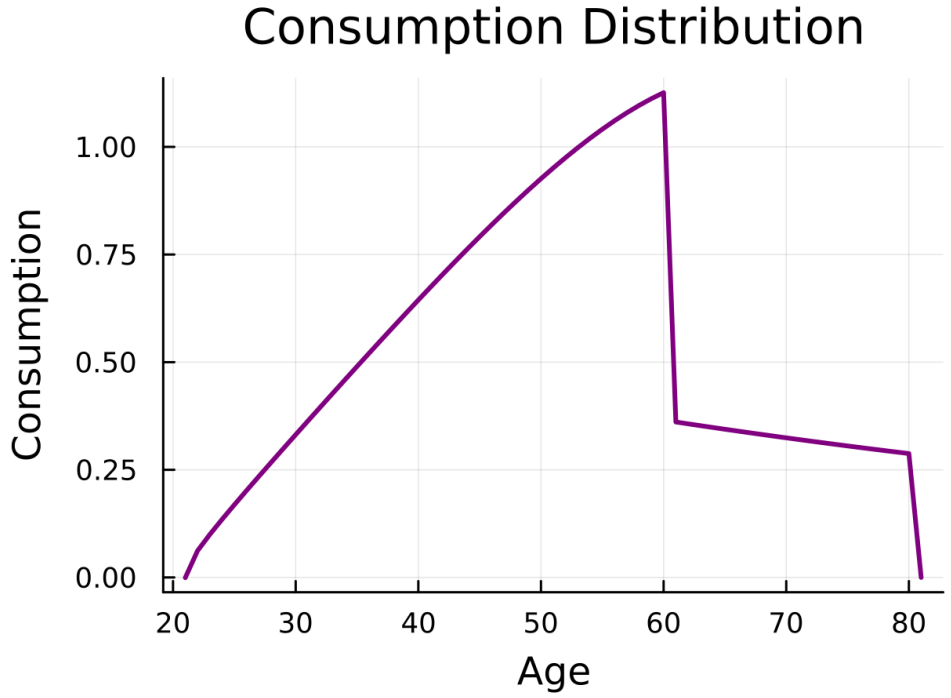

Capital Accumulation Over Life Cycle
Capital
Age
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
1
2
3
4
5
6


Labor Supply Over Life Cycle
Labor
Age
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
1
2
3
4
5
6


Capital Distribution
Capital
Age
0
1
2
3
20
30
40
50
60
70
80


Labor Distribution
Labor
Age
0.0
0.1
0.2
0.3
0.4
20
30
40
50
60
70
80


Income Distribution
Wage
Age
0
1
2
3
20
30
40
50
60
70
80


Consumption Distribution
Consumption
Age
0.00
0.25
0.50
0.75
1.00
20
30
40
50
60
70
80